

C3-091 Practice Pack

Attempt-first independent practice | Reviews locked

BOOK	->	QUALITY	->	OPERATE	->	PROVE
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2026 LEARNER EDITION

Option B institute family: Charcoal #1F2937 | Teal #14B8A6 | Soft Blue #3B82F6 | AMOLED-ready companion shell.

Practice operating rule

Attempt each Pxx only after studying the matching lesson. Save work under evidence/Lxx/attempt_001/. A passing script/test alone is never enough: include the artifact, observed result, independent validation, confidence, uncertainty and one sanitized error/recovery note.

Starter: Practice/C3_091_PRACTICE_STARTER_PROJECT_RC1.zip

P01 - Data-quality dimensions and enforceable contracts

Matches C3-08-L01. Keep R01 closed until this attempt exists.

Your independent job

Write the orders contract: grain/key/schema/status domain/owners and at least four quality rules with severity and publish action.

Required evidence

1) your actual artifact/code/config/analysis; 2) observed command/result or honest runtime boundary; 3) an independent validation/negative test; 4) confidence 1-5 + biggest uncertainty; 5) explicit owner/action/recovery boundary for incident/release/security work; 6) changed transfer after review only if needed.

Pass condition

Correct, reproducible, independently validated and explainable on changed conditions. A green task or passing test alone is insufficient.

P02 - Blocking vs warning checks and severity design

Matches C3-08-L02. Keep R02 closed until this attempt exists.

Your independent job

Classify duplicate key, missing customer_id and invalid status as block/warn/info from consumer impact; define owner/escalation.

Required evidence

1) your actual artifact/code/config/analysis; 2) observed command/result or honest runtime boundary; 3) an independent validation/negative test; 4) confidence 1-5 + biggest uncertainty; 5) explicit owner/action/recovery boundary for incident/release/security work; 6) changed transfer after review only if needed.

Pass condition

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P03 - Source-to-target reconciliation patterns

Matches C3-08-L03. Keep R03 closed until this attempt exists.

Your independent job

Run reconciliation. Identify missing O506, mismatched O503 amount and explain the total delta. Then design the offsetting-error transfer.

Required evidence

1) your actual artifact/code/config/analysis; 2) observed command/result or honest runtime boundary; 3) an independent validation/negative test; 4) confidence 1-5 + biggest uncertainty; 5) explicit owner/action/recovery boundary for incident/release/security work; 6) changed transfer after review only if needed.

Pass condition

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P04 - Freshness, completeness and timeliness controls

Matches C3-08-L04. Keep R04 closed until this attempt exists.

Your independent job

Using the fixed observation time, compute freshness and expected partition completeness; state breach and alert behavior.

Required evidence

1) your actual artifact/code/config/analysis; 2) observed command/result or honest runtime boundary; 3) an independent validation/negative test; 4) confidence 1-5 + biggest uncertainty; 5) explicit owner/action/recovery boundary for incident/release/security work; 6) changed transfer after review only if needed.

Pass condition

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P05 - Metadata, lineage and ownership

Matches C3-08-L05. Keep R05 closed until this attempt exists.

Your independent job

Create an OpenLineage-shaped run/job/dataset metadata record with owners/code version and downstream blast-radius note.

Required evidence

1) your actual artifact/code/config/analysis; 2) observed command/result or honest runtime boundary; 3) an independent validation/negative test; 4) confidence 1-5 + biggest uncertainty; 5) explicit owner/action/recovery boundary for incident/release/security work; 6) changed transfer after review only if needed.

Pass condition

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P06 - Logs, metrics and tracing concepts

Matches C3-08-L06. Keep R06 closed until this attempt exists.

Your independent job

Design a signal contract joining logs/metrics/trace context by run_id/trace_id without raw PII/secrets.

Required evidence

1) your actual artifact/code/config/analysis; 2) observed command/result or honest runtime boundary; 3) an independent validation/negative test; 4) confidence 1-5 + biggest uncertainty; 5) explicit owner/action/recovery boundary for incident/release/security work; 6) changed transfer after review only if needed.

Pass condition

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P07 - Pipeline SLIs, SLOs and alert design

Matches C3-08-L07. Keep R07 closed until this attempt exists.

Your independent job

Compute validated-publish and on-time SLIs from history; define SLO, page/ticket thresholds and owner action.

Required evidence

1) your actual artifact/code/config/analysis; 2) observed command/result or honest runtime boundary; 3) an independent validation/negative test; 4) confidence 1-5 + biggest uncertainty; 5) explicit owner/action/recovery boundary for incident/release/security work; 6) changed transfer after review only if needed.

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P08 - Incident triage and root-cause analysis

Matches C3-08-L08. Keep R08 closed until this attempt exists.

Your independent job

Build the incident timeline/RCA from the practice evidence. Include containment, repair scope, post-repair reconcile and prevention.

Required evidence

1) your actual artifact/code/config/analysis; 2) observed command/result or honest runtime boundary; 3) an independent validation/negative test; 4) confidence 1-5 + biggest uncertainty; 5) explicit owner/action/recovery boundary for incident/release/security work; 6) changed transfer after review only if needed.

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P09 - Runbooks and operational handoff

Matches C3-08-L09. Keep R09 closed until this attempt exists.

Your independent job

Complete an operator runbook for the practice harness, then add the changed expired-credential failure branch.

Required evidence

1) your actual artifact/code/config/analysis; 2) observed command/result or honest runtime boundary; 3) an independent validation/negative test; 4) confidence 1-5 + biggest uncertainty; 5) explicit owner/action/recovery boundary for incident/release/security work; 6) changed transfer after review only if needed.

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P10 - Secrets, identity and access-control engineering

Matches C3-08-L10. Keep R10 closed until this attempt exists.

Your independent job

Design least-privilege CI/deployment access. Show minimal workflow permissions, protected production environment and OIDC boundary.

Required evidence

1) your actual artifact/code/config/analysis; 2) observed command/result or honest runtime boundary; 3) an independent validation/negative test; 4) confidence 1-5 + biggest uncertainty; 5) explicit owner/action/recovery boundary for incident/release/security work; 6) changed transfer after review only if needed.

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P11 - Encryption and sensitive-data handling

Matches C3-08-L11. Keep R11 closed until this attempt exists.

Your independent job

Sanitize the sample evidence and write classification/encryption/access/masking/retention rules.

Required evidence

1) your actual artifact/code/config/analysis; 2) observed command/result or honest runtime boundary; 3) an independent validation/negative test; 4) confidence 1-5 + biggest uncertainty; 5) explicit owner/action/recovery boundary for incident/release/security work; 6) changed transfer after review only if needed.

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P12 - CI/CD for data projects and automated checks

Matches C3-08-L12. Keep R12 closed until this attempt exists.

Your independent job

Complete the CI workflow/release checklist so unsafe tests cannot deploy and production secrets are not exposed to PR jobs.

Required evidence

1) your actual artifact/code/config/analysis; 2) observed command/result or honest runtime boundary; 3) an independent validation/negative test; 4) confidence 1-5 + biggest uncertainty; 5) explicit owner/action/recovery boundary for incident/release/security work; 6) changed transfer after review only if needed.

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P13 - Environment promotion and rollback

Matches C3-08-L13. Keep R13 closed until this attempt exists.

Your independent job

Decide rollback vs roll-forward for the state-changing release case and define post-deploy smoke/reconciliation.

Required evidence

1) your actual artifact/code/config/analysis; 2) observed command/result or honest runtime boundary; 3) an independent validation/negative test; 4) confidence 1-5 + biggest uncertainty; 5) explicit owner/action/recovery boundary for incident/release/security work; 6) changed transfer after review only if needed.

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P14 - Infrastructure-as-code concepts for data systems

Matches C3-08-L14. Keep R14 closed until this attempt exists.

Your independent job

Review the learning-only Terraform file. Explain plan/state/locking/drift; do not provision paid infrastructure.

Required evidence

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P15 - Cost, performance and reliability trade-offs

Matches C3-08-L15. Keep R15 closed until this attempt exists.

Your independent job

Write an ADR comparing per-run full reconciliation vs sampled checks, preserving correctness/security floors and a review trigger.

Required evidence

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